

Cost of Living (US) Data Visualization

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Data 380

Data Analysis:

1. Cleaning Data

case_id	state	isMetro	areaname	county	family_member_count	housing_cost	food_cost	transportation_cost	healthcare_cost	other_necessities_cost
1	AL	True	Montgomery, AL MSA	Autauga County	1p0c	8505.72876	3454.91712	10829.16876	5737.47984	4333.81344
1	AL	True	Montgomery, AL MSA	Autauga County	1p1c	12067.50240	5091.70788	11588.19288	8659.55640	6217.45896
1	AL	True	Montgomery, AL MSA	Autauga County	1p2c	12067.50240	7460.20308	12361.77720	11581.63260	7075.65816
1	AL	True	Montgomery, AL MSA	Autauga County	1p3c	15257.15040	9952.23924	13452.18600	14503.70760	9134.35620
1	AL	True	Montgomery, AL MSA	Autauga County	1p4c	15257.15040	12182.21400	13744.59840	17425.78560	9942.38396

childcare_cost	taxes	total_cost	median_family_income
0.0000	6392.94504	39254.0532	73010.414062
6147.8298	7422.07836	57194.3256	73010.414062
15824.6940	9769.56228	76141.0308	73010.414062
18802.1892	13101.70320	94203.5328	73010.414062
18802.1892	13469.21880	100823.5200	73010.414062

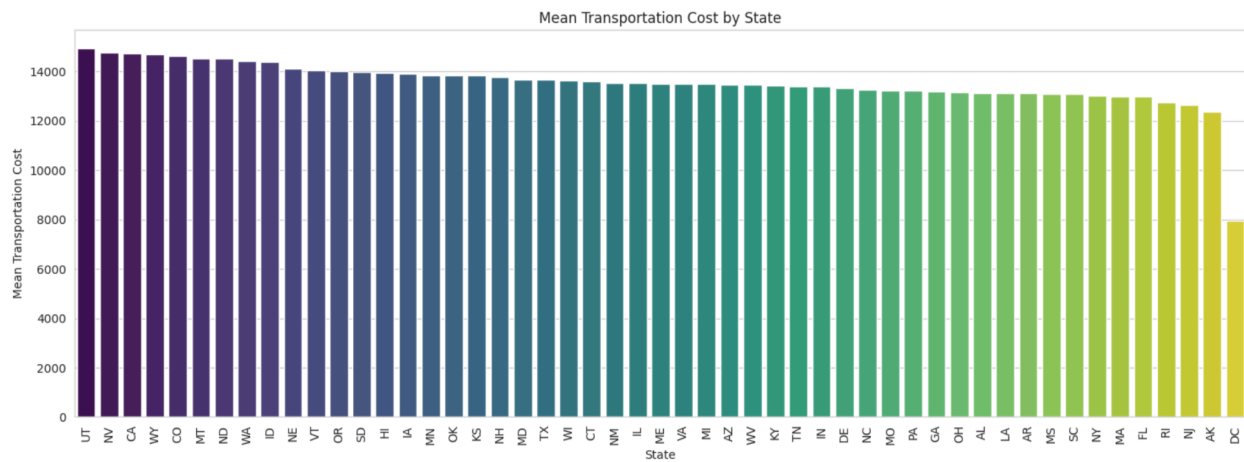
Kept all data because all is necessary for analysis.

2. Interactive Table

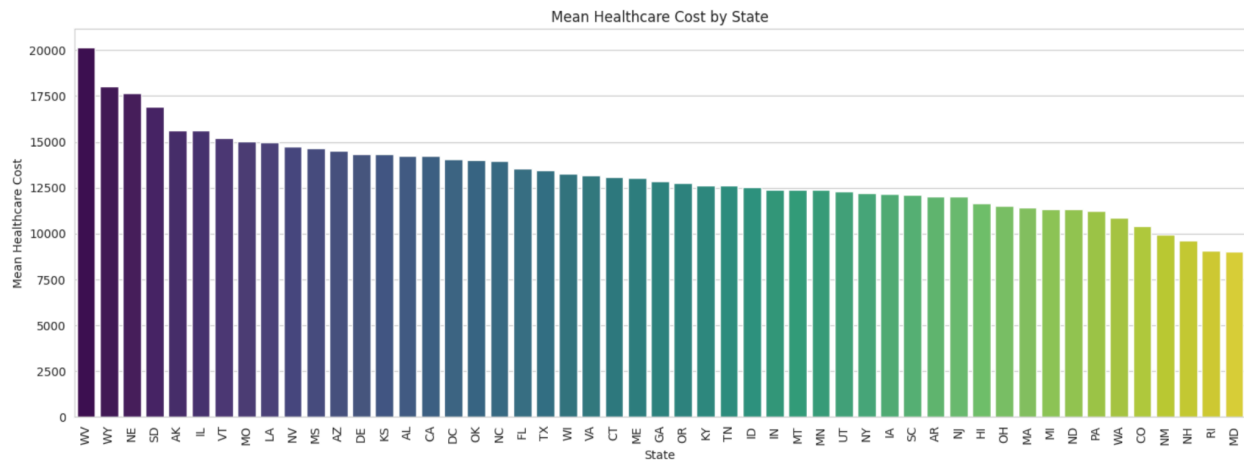
10	entries per page									Search:
state	housing_cost	food_cost	transportation_cost	healthcare_cost	other_necessities_cost	childcare_cost	taxes	total_cost	median_family_income	
AK	15520.43	8694.78	12361.72	15641.82	8774.13	15194.39	9047.29	85234.56	81242.83	
AL	9181.39	8480.71	13141.57	14236.29	6399.68	9552.57	7975.35	68967.57	58099.87	
AR	8886.49	7788.49	13131.93	12038.51	6042.00	7732.10	5872.37	61491.89	55025.57	
AZ	12793.91	8146.29	13479.58	14532.48	7587.46	11314.89	8131.32	75985.94	59661.88	
CA	20232.49	9673.95	14740.17	14221.49	10836.28	10761.33	12713.44	93179.16	82922.17	
CO	14133.44	9168.69	14616.03	10427.42	8443.28	11290.98	9515.95	77595.78	75158.81	
CT	16953.56	9703.83	13606.71	13085.25	9659.02	13624.39	13380.73	90013.50	103228.23	
DC	21089.39	11631.16	7961.97	14065.90	11855.94	31940.49	21744.79	120289.64	116607.39	
DE	14449.79	8989.75	13320.45	14330.67	8493.07	11319.14	10602.65	81505.52	81198.66	
FL	13166.64	8955.00	12972.21	13543.69	8015.54	8797.53	6071.10	71521.72	64294.18	

3.

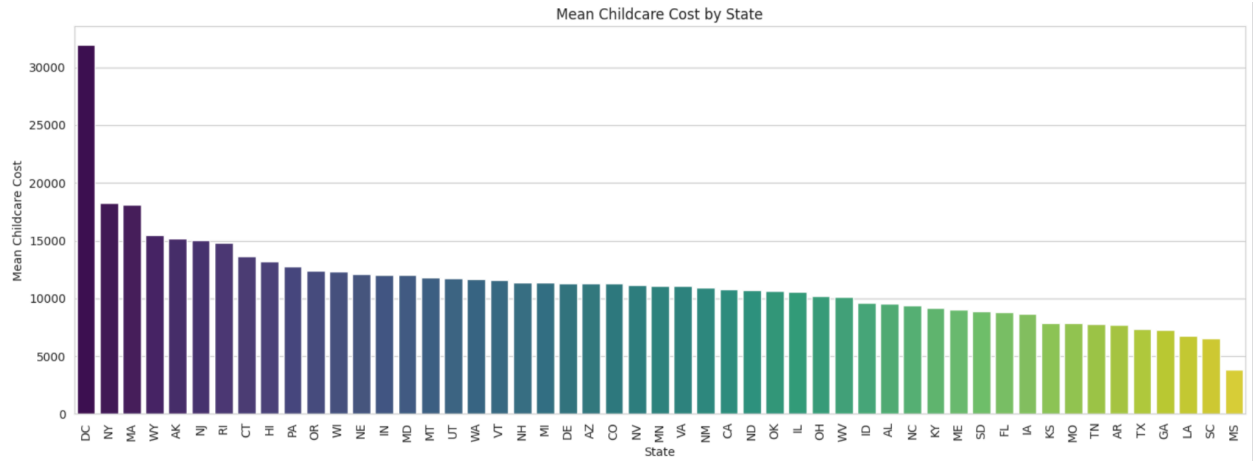
Distribution of Cost Components and Income by State



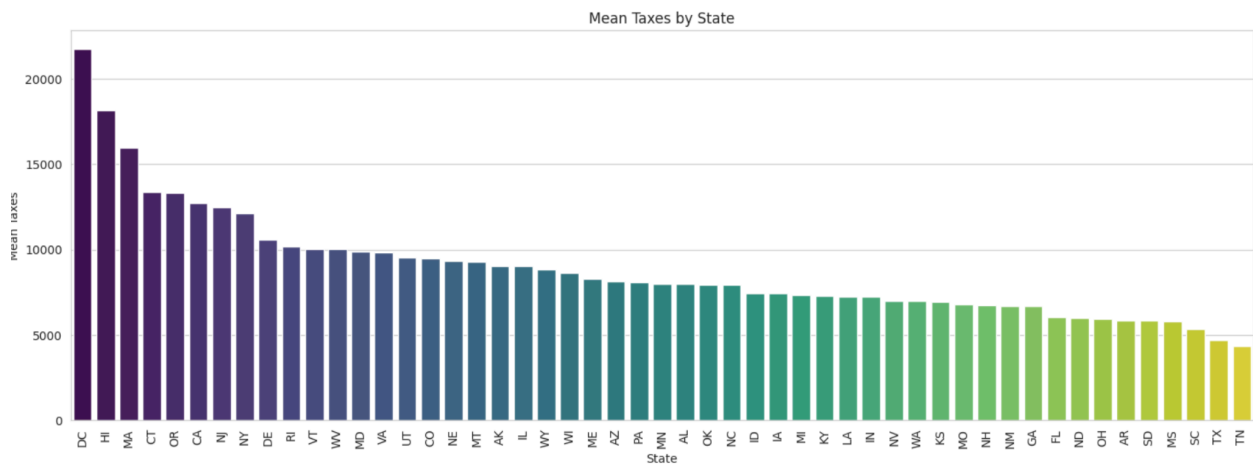
Many western state's transportation costs are higher on average, but overall, the range is quite small, all states except one are within \$2000 of each other. DC is an outlier with just under \$8000.



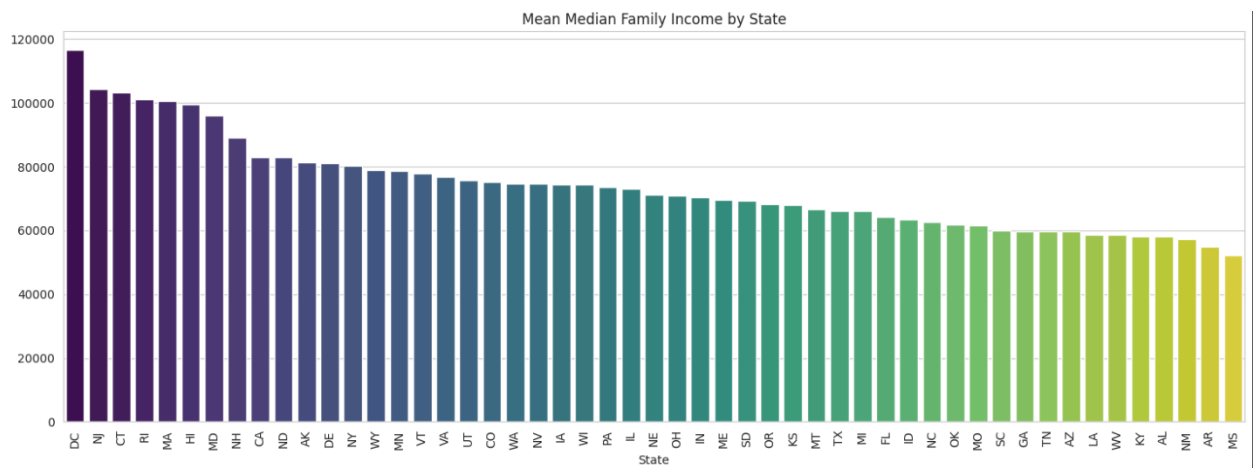
There is no geographical correlation for mean healthcare cost. West Virginia is an outlier, with their mean healthcare cost being ~ \$2500 higher than every other state.



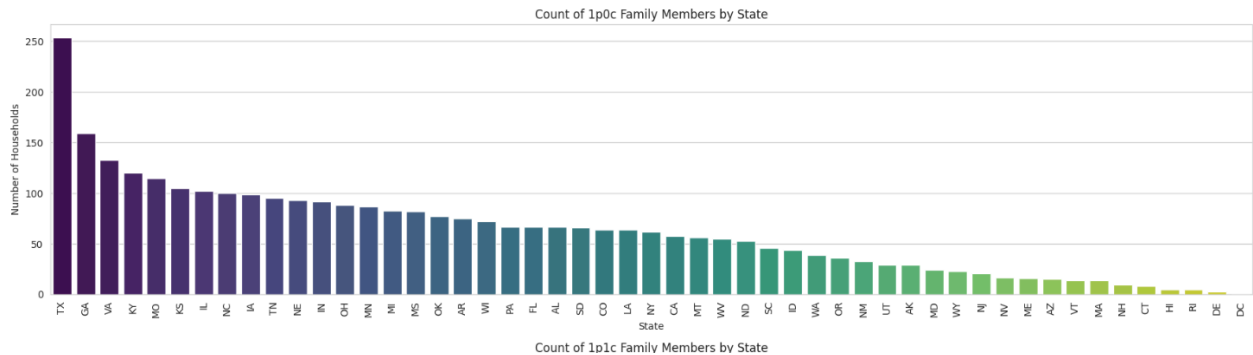
There isn't a true geographical correlation, but many northeastern states find themselves in the top half of this graph. DC is a clear outlier being \$13000+ higher than any other state.



There is no geographical correlation. DC is an outlier.



Many coastal states are higher up in median family income. DC is the highest at just under \$120,000.



All of the family member type graphs look the same because in the dataset, each state made sure to have the same number of each family type.

4.

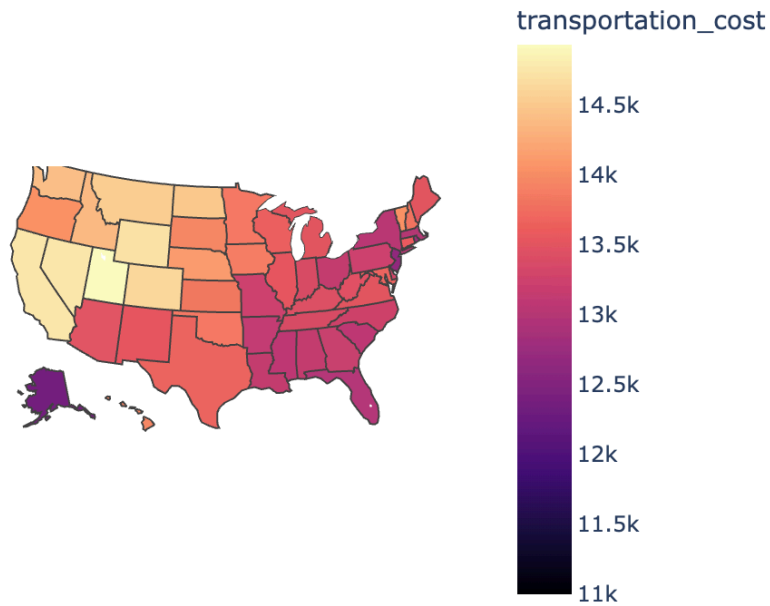
state	housing_cost	food_cost	transportation_cost	healthcare_cost	other_necessities_cost
AK	15520.430000	8694.780000	12361.720000	15641.820000	8774.130000
AL	9181.390000	8480.710000	13141.570000	14236.290000	6399.680000
AR	8886.490000	7788.490000	13131.930000	12038.510000	6042.000000
AZ	12793.910000	8146.290000	13479.580000	14532.480000	7587.460000
CA	20232.490000	9673.950000	14740.170000	14221.490000	10836.280000
CO	14133.440000	9168.690000	14616.030000	10427.420000	8443.280000
CT	16953.560000	9703.830000	13606.710000	13085.250000	9659.020000
DC	21089.390000	11631.160000	7961.970000	14065.900000	11855.940000
DE	14449.790000	8989.750000	13320.450000	14330.670000	8493.070000
FL	13166.640000	8955.000000	12972.210000	13543.690000	8015.540000

childcare_cost	taxes	total_cost	median_family_income
15194.390000	9047.290000	85234.560000	81242.830000
9552.570000	7975.350000	68967.570000	58099.870000
7732.100000	5872.370000	61491.890000	55025.570000
11314.890000	8131.320000	75985.940000	59661.880000
10761.330000	12713.440000	93179.160000	82922.170000
11290.980000	9515.950000	77595.780000	75158.810000
13624.390000	13380.730000	90013.500000	103228.230000
31940.490000	21744.790000	120289.640000	116607.390000
11319.140000	10602.650000	81505.520000	81198.660000
8797.530000	6071.100000	71521.720000	64294.180000

Interactive table using colors. The lighter the color is, the higher the number it represents.

5.

Transportation Cost by State (Choropleth Map)



Map showing transportation cost using Choropleth. Lighter color represent higher cost.

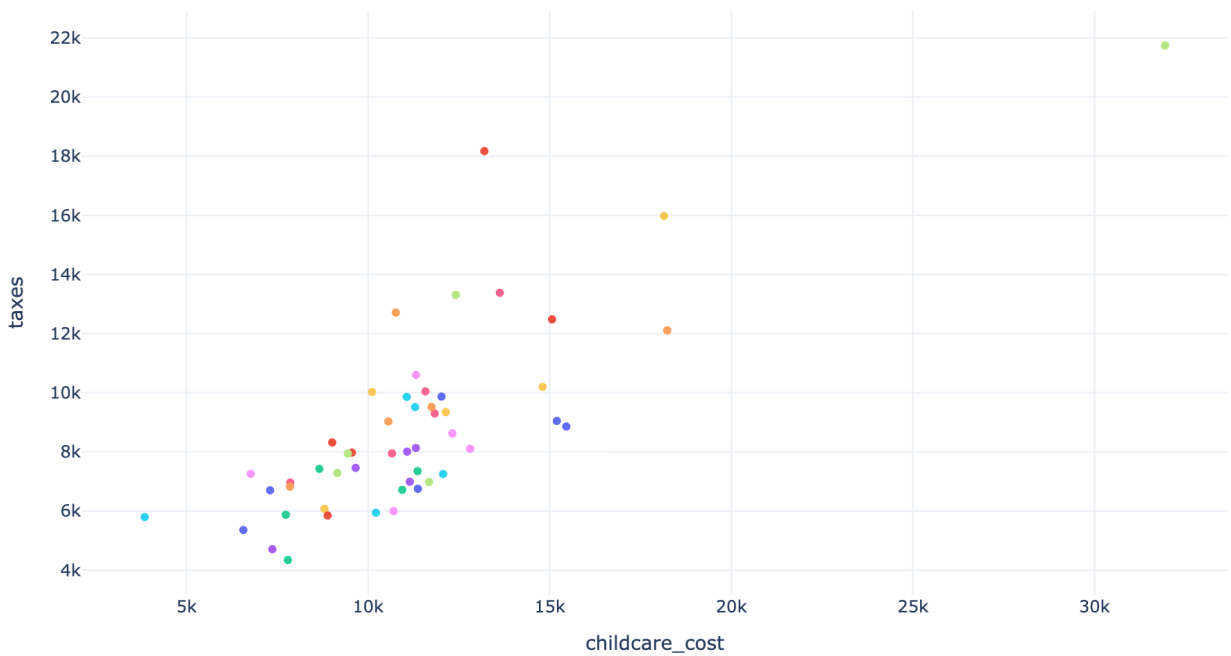
6.

Transportation Cost vs. Total Cost by State



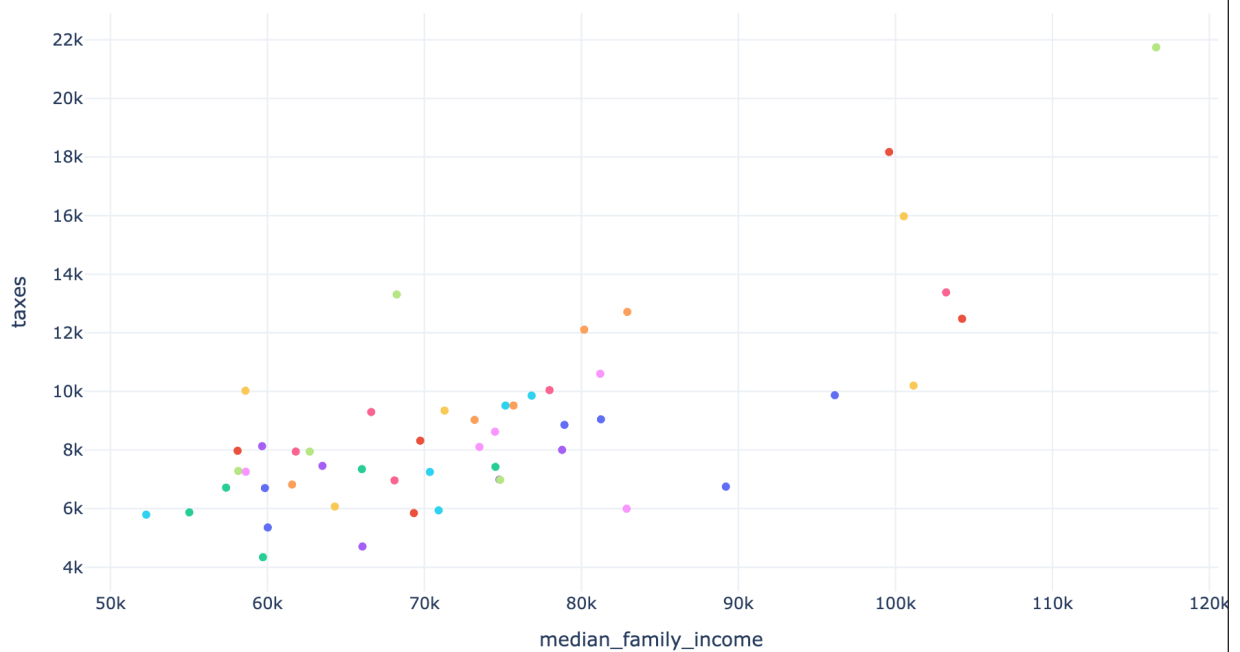
Scatter plot showing transportation cost vs total cost. Many reside in the top-left quadrant. DC is a major outlier with an extremely high total cost, but also an extremely low transportation cost.

Taxes vs. Childcare Cost by State

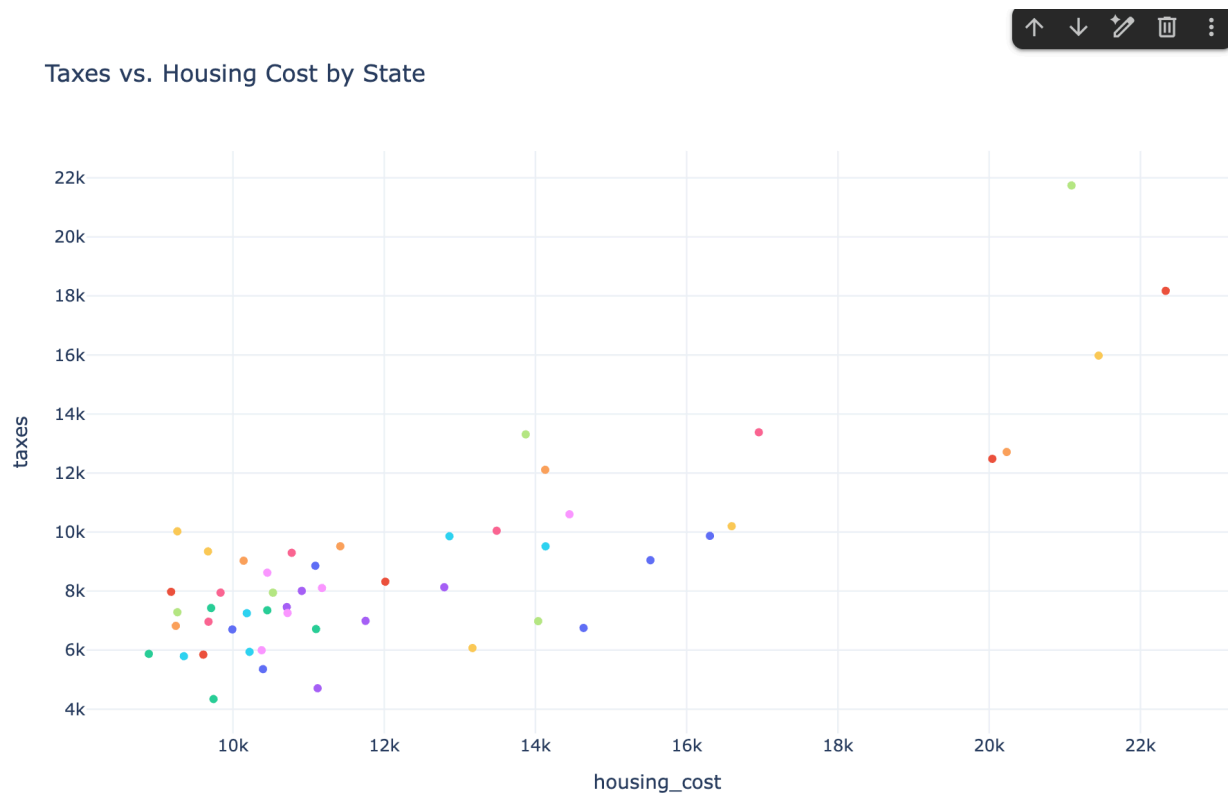


Scatter plot showing Taxes vs Childcare Cost. There is a strong positive correlation. DC is an outlier with very high taxes and childcare cost

Taxes vs. Median Family Income by State



Scatter plot showing Taxes vs Median Family Income. There is a strong positive correlation. DC is the highest with almost \$22k in taxes and about \$117k median family income.



Scatter plot showing Taxes vs Housing cost. There is a strong positive correlation. There are no outliers.

7.

Top 12 Most Expensive Counties for Childcare Cost:

10 entries per page Search:

county	childcare_cost
493 District of Columbia	31940.492400
1187 New York County	29620.246800
1379 Queens County	28050.334320
1439 Rockland County	27260.377200
1796 Westchester County	26661.030720
1617 Suffolk County	25288.633680
206 Bronx County	24340.542960
1203 Norfolk County	23606.173680
519 Dutchess County	22844.859600
1714 Ulster County	21947.080944

Showing 1 to 10 of 12 entries

Top 12 Most Expensive Counties for Childcare Cost

8.

Top 12 Counties with Highest Taxes:

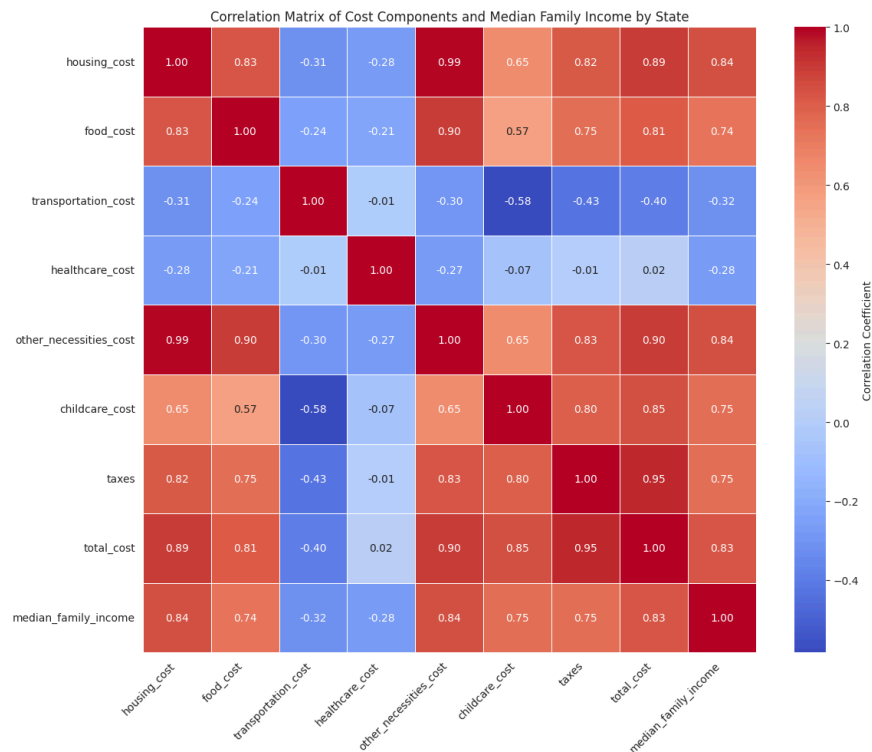
10 entries per page Search:

	county	taxes
1480	San Mateo County	34738.964880
1043	Marin County	31637.529360
1475	San Francisco County	27174.135960
1187	New York County	26517.054360
1492	Santa Clara County	24651.857760
1439	Rockland County	23463.096600
1379	Queens County	22807.598208
1203	Norfolk County	22347.509448
1796	Westchester County	22292.750676
1617	Suffolk County	22256.451666

Showing 1 to 10 of 12 entries

Top 12 counties with the highest taxes.

9.



Correlation Matrix of Cost Components and Median Family Income by state. Housing cost and other necessities cost have a very strong correlation. Healthcare cost and other necessities cost have a weak correlation.

10.



Regression line of Total Cost vs Taxes by state. Strong positive correlation.

11.

```
... Q1 (25th percentile) of food_cost: 8129.10
    Q3 (75th percentile) of food_cost: 9054.14
    IQR (Interquartile Range) of food_cost: 925.03
    Lower Bound for Outliers: 6741.56
    Upper Bound for Outliers: 10441.68
```

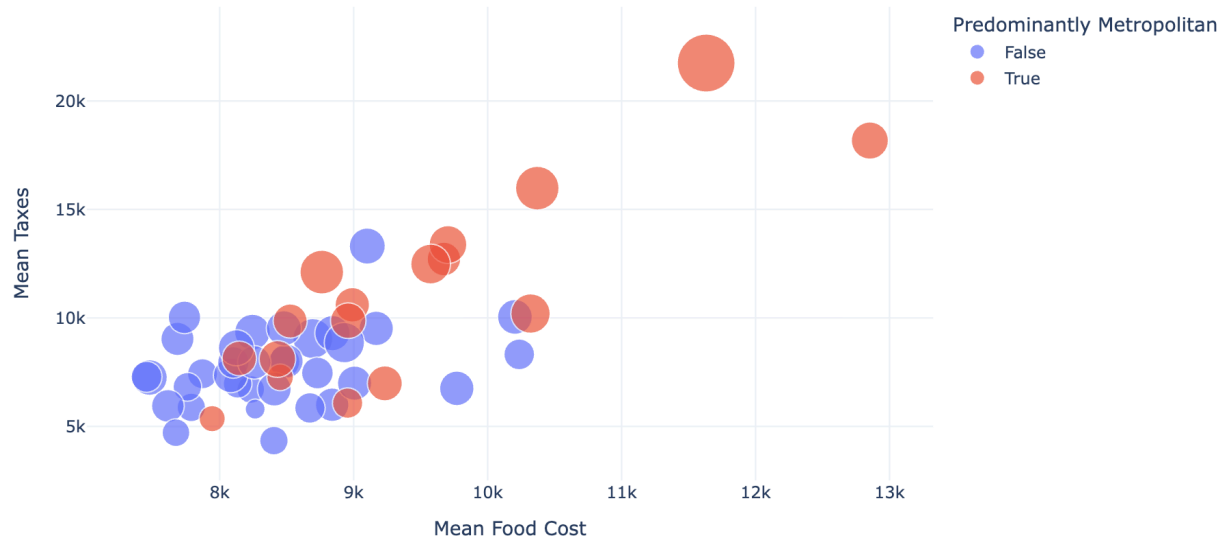
States identified as outliers for food_cost:

state	food_cost
7 DC	11631.16
11 HI	12852.93

DC and Hawaii are outliers for food cost.

12.

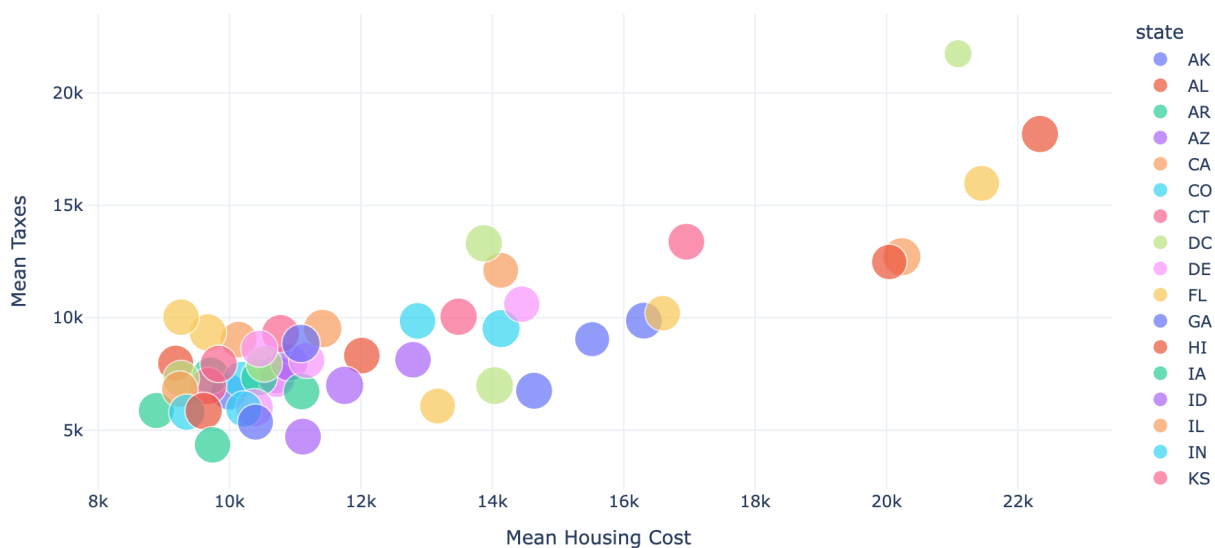
Food Cost vs. Taxes by State (Size: Childcare Cost, Color: isMetro)



Bubble plot showing states based on mean taxes, mean food cost, metropolitan status, and childcare cost. Red bubbles are predominantly metropolitan, blue are not. The bubble size represents childcare cost (bigger bubble = higher cost). Strong positive correlation between mean taxes and mean food cost.

13.

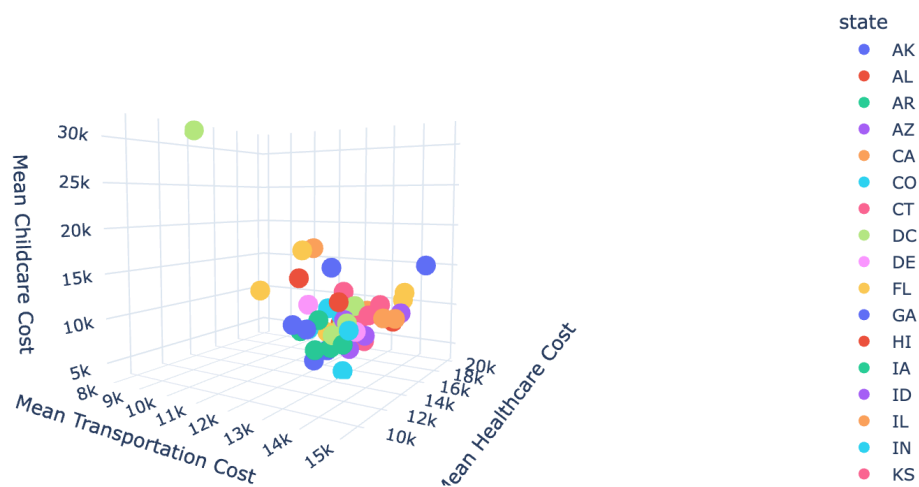
Housing Cost vs. Taxes by State (Size: Transportation Cost, Color: State)



Bubble plot showing states based on mean taxes, mean housing cost, transportation cost, and state based on color. Bigger bubble = higher transportation cost. Strong positive correlation between mean taxes and mean housing cost.

14.

3D Plot of Transportation, Healthcare, and Childcare Costs by State

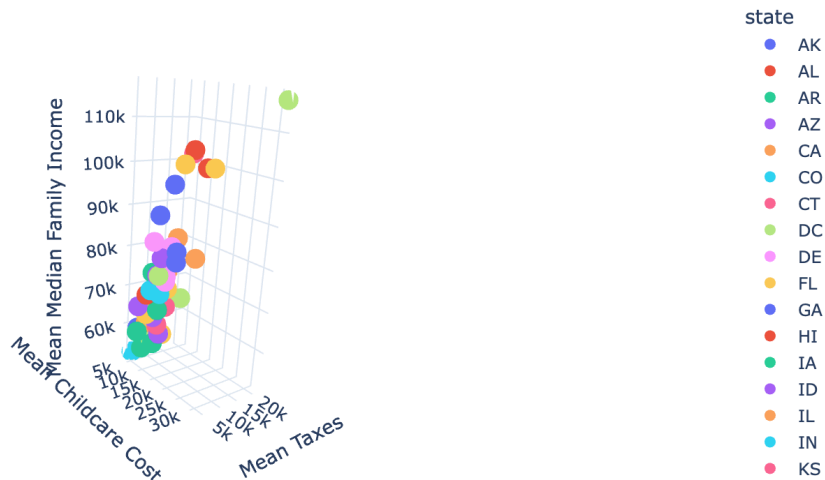


3D plot showing states by color and based on mean childcare cost, mean transportation cost, and mean healthcare cost. The majority of states are grouped together, but DC is an outlier with extremely high childcare and extremely low mean transportation cost.

15.



3D Plot of Childcare Cost, Taxes, and Median Family Income by State



3D plot showing states by color and based on mean family income, mean childcare cost, and mean taxes. There is a strong positive correlation between the three variables. DC has a much higher mean childcare cost and mean family income.

16.

Explained variance ratio by each principal component:
[0.67458086 0.13547299 0.09634518]
Total explained variance: 0.91

First 5 rows of the PCA components DataFrame:

PC1	PC2	PC3	state
1.551379	1.463528	-0.119419	AK
-1.624223	0.791863	-0.242032	AL
-2.624577	-0.064480	-1.094172	AR
-0.686146	0.665555	0.206533	AZ
2.858522	-0.861238	2.182109	CA

The first 3 principal components capture 91% of the total variability in the cost and income across states.

17.

... First 5 rows of the NMF components DataFrame:

	NMF1	NMF2	NMF3	state
	142.042824	122.728444	132.677957	AK
	139.975447	139.706573	37.020204	AL
	135.519172	105.317929	31.571878	AR
	136.333265	173.497598	50.628185	AZ
	149.848850	157.928654	130.712080	CA

Original features' weights in NMF components (components_ attribute of NMF):

	housing_cost	food_cost	transportation_cost	healthcare_cost	other_necessities_cost	childcare_cost	taxes
NMF1	37.616829	41.448942	87.173660	71.106308	28.648662	15.460188	5.836360
NMF2	21.641193	11.234751	5.719278	30.812137	11.912235	35.694460	37.858354
NMF3	49.537500	17.178425	4.833218	0.000000	24.173801	54.767869	42.543423

NMF of the cost of living dataset.

Conclusion:

There are clearly many trends in the Cost of Living (US) dataset. First, there are multiple geographical correlations when it comes to this dataset. Transportation cost seems to be more expensive in western states, with some outliers obviously. Many states in the northeastern US have higher childcare costs than any other region in the US. A trend with median family income is that coastal states were taking up the majority of the top spots. In the Choropleth map showing transportation cost, the trend of western states being more expensive is more clear. It almost seems like the more northwest a state is, the higher their transportation cost is. Many of the most expensive childcare counties are in New York and surrounding states. Many of the highest taxed counties are in California and New York. There are a lot of geographical trends in this dataset.

Many scatter and bubble plots show strong positive correlations between different variables. There are strong positive correlations between taxes and childcare cost, taxes and median family income, taxes and housing cost, taxes and total cost, and lastly taxes and food cost. Clearly the higher the taxes are for a state, the higher their other costs and income will be as well. In the 3D plots, there are large groupings of states. This shows that many states may have differences in two variables, but they are much closer to each other on a third variable. There are many positive correlations throughout this dataset.

Overall, after analyzing this dataset, it is clear that there are many trends throughout the US when it comes to cost of living and other similar variables.

Appendix:

1. Gemini used for cleaning.

```
import os
import pandas as pd # Ensure pandas is imported

file_path = '/content/cost_of_living_us.csv'

if os.path.exists(file_path):
    print(f"File '{file_path}' found. Attempting to load DataFrame...")
    try:
        df = pd.read_csv(file_path)
        print("DataFrame 'df' loaded successfully.")
        display(df.head())
    except Exception as e:
        print(f"Error loading DataFrame: {e}")
else:
    print(f"Error: File '{file_path}' not found at the specified path.
Please ensure it is correctly uploaded and accessible.")

print(f"Shape of DataFrame before dropping nulls: {df.shape}")

df_cleaned = df.dropna()

print(f"Shape of DataFrame after dropping nulls: {df_cleaned.shape}")

display(df_cleaned.head())
```

2. Gemini used for interactive table creation.

```
# Install itables for interactive table display
%pip install itables

from itables import init_notebook_mode, show
init_notebook_mode(all_interactive=True)

# Select relevant cost columns for summarization
cost_columns = [
    'housing_cost',
    'food_cost',
    'transportation_cost',
```

```

    'healthcare_cost',
    'other_necessities_cost',
    'childcare_cost',
    'taxes',
    'total_cost',
    'median_family_income'
]

# Group by 'state' and calculate the mean of cost columns
df_summary =
df_cleaned.groupby('state')[cost_columns].mean().reset_index()

# Round the values to 2 decimal places for better readability
df_summary = df_summary.round(2)

# Display the interactive summary table
show(df_summary, scrollX=True, paging=True)

```

3. Gemini used for bar graph creation.

```

import matplotlib.pyplot as plt
import seaborn as sns

# Set a consistent style for the plots
sns.set_style("whitegrid")

# List of columns to plot from df_summary
plot_columns_summary = [
    'transportation_cost',
    'healthcare_cost',
    'childcare_cost',
    'taxes',
    'median_family_income'
]

# Create a figure to hold all subplots
fig, axes = plt.subplots(len(plot_columns_summary), 1, figsize=(15, 6 *
len(plot_columns_summary)))
fig.suptitle('Distribution of Cost Components and Income by State',
y=0.99, fontsize=16)

```



```

for i, col in enumerate(plot_columns_summary):
    ax = axes[i]
    sns.barplot(x='state', y=col, data=df_summary.sort_values(by=col,
ascending=False), palette='viridis', ax=ax)
    ax.set_title(f'Mean {col.replace("_", " ").title()} by State')
    ax.set_xlabel('State')
    ax.set_ylabel(f'Mean {col.replace("_", " ").title()}')
    ax.tick_params(axis='x', rotation=90)

plt.tight_layout(rect=[0, 0.03, 1, 0.96]) # Adjust layout to prevent title
overlap
plt.show()

```

4. Gemini used for table creation.

```

from itables import options

# Define the columns to apply styling to (all numeric columns except
'state')
style_columns = df_summary.select_dtypes(include=['number']).columns

# Set itables option to allow HTML rendering for Styler objects
options.allow_html = True

# Create a Styler object and apply background gradients
styler = df_summary.style.background_gradient(cmap='viridis',
subset=style_columns)

# Display the styled table
display(styler)

```

5. Gemini used for Choropleth graph creation.

```

import plotly.express as px

# Get the maximum transportation cost to define the upper bound for the
color scale
max_transportation_cost = df_summary['transportation_cost'].max()

fig = px.choropleth(
    df_summary,
    locations='state',

```

```

locationmode='USA-states',
color='transportation_cost',
scope='usa',
color_continuous_scale='Magma',
range_color=[11000, max_transportation_cost], # Set custom range to
start from 11k
title='Transportation Cost by State (Choropleth Map)',
hover_name='state',
hover_data={'transportation_cost': True}
)

fig.show()

```

6. Gemini used for scatter plot creation.

```

import plotly.express as px

# Define the pairs of columns to plot
plot_pairs = [
    ('total_cost', 'transportation_cost'),
    ('childcare_cost', 'taxes'),
    ('median_family_income', 'taxes'),
    ('housing_cost', 'taxes')
]

for i, (x_col, y_col) in enumerate(plot_pairs):
    fig = px.scatter(
        df_summary,
        x=x_col,
        y=y_col,
        color='state', # Use 'state' for coloring
        title=f'{y_col.replace("_", " ").title()} vs. {x_col.replace("_", " ").title()} by State',
        width=1000, # Make it wider for better readability
        height=600, # Adjust height as needed
        hover_name='state', # Show state name on hover
        template='plotly_white' # Use a clean template
    )
    fig.show()

```

7. Gemini used for sheet creation.

```

childcare_cost_by_county =
df_cleaned.groupby('county')['childcare_cost'].mean().reset_index()

top_12_childcare_counties =
childcare_cost_by_county.sort_values(by='childcare_cost',
ascending=False).head(12)

print("Top 12 Most Expensive Counties for Childcare Cost:")
display(top_12_childcare_counties)

```

8. Gemini used for sheet creation.

```

taxes_by_county =
df_cleaned.groupby('county')['taxes'].mean().reset_index()

top_12_tax_counties = taxes_by_county.sort_values(by='taxes',
ascending=False).head(12)

print("Top 12 Counties with Highest Taxes:")
display(top_12_tax_counties)

```

9. Gemini used for correlation matrix creation.

```

import seaborn as sns
import matplotlib.pyplot as plt

# Select the relevant numerical columns for correlation from df_summary
# 'state' is not a numerical column, so it's excluded by default with
.corr()
correlation_matrix = df_summary[cost_columns].corr()

plt.figure(figsize=(12, 10))
sns.heatmap(
    correlation_matrix,
    annot=True, # Show the correlation values on the heatmap
    cmap='coolwarm', # Choose a diverging colormap
    fmt=".2f", # Format annotation values to 2 decimal places
    linewidths=.5, # Add lines between cells
    cbar_kws={'label': 'Correlation Coefficient'}
)
plt.title('Correlation Matrix of Cost Components and Median Family Income
by State')

```

```
plt.xticks(rotation=45, ha='right')
plt.yticks(rotation=0)
plt.tight_layout()
plt.show()
```

10. Gemini used for regression line creation.

```
import seaborn as sns
import matplotlib.pyplot as plt

# Create a scatter plot with a regression line
plt.figure(figsize=(10, 7))
sns.regplot(x='total_cost', y='taxes', data=df_summary,
            scatter_kws={'alpha':0.6}, line_kws={'color':'red'})
plt.title('Regression Analysis: Total Cost vs. Taxes by State')
plt.xlabel('Total Cost')
plt.ylabel('Taxes')
plt.grid(True)
plt.show()
```

11. Gemini used for the outlier equation.

```
import numpy as np

# Calculate Q1 (25th percentile) and Q3 (75th percentile) for food_cost
Q1 = df_summary['food_cost'].quantile(0.25)
Q3 = df_summary['food_cost'].quantile(0.75)

# Calculate IQR (Interquartile Range)
IQR = Q3 - Q1

# Define outlier bounds
lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR

# Identify outliers
outliers_food_cost = df_summary[(df_summary['food_cost'] < lower_bound) |
                                 (df_summary['food_cost'] > upper_bound)]

print(f"Q1 (25th percentile) of food_cost: {Q1:.2f}")
print(f"Q3 (75th percentile) of food_cost: {Q3:.2f}")
print(f"IQR (Interquartile Range) of food_cost: {IQR:.2f}")
```

```

print(f"Lower Bound for Outliers: {lower_bound:.2f}")
print(f"Upper Bound for Outliers: {upper_bound:.2f}")

print("\nStates identified as outliers for food_cost:")
display(outliers_food_cost[['state',
'food_cost']].sort_values(by='food_cost'))

```

12. Gemini used for plot creation.

```

# Calculate the mode of 'isMetro' for each state
is_metro_by_state = df_cleaned.groupby('state')['isMetro'].agg(lambda x:
x.mode()[0]).reset_index()
is_metro_by_state.rename(columns={'isMetro': 'predominant_isMetro'},
inplace=True)

# Merge this information into df_summary
df_summary_metro = pd.merge(df_summary, is_metro_by_state, on='state',
how='left')

display(df_summary_metro.head())

```

```

import plotly.express as px

fig = px.scatter(
    df_summary_metro,
    x='food_cost',
    y='taxes',
    size='childcare_cost',
    color='predominant_isMetro', # Color based on whether the state is
predominantly metro or not
    hover_name='state',
    log_x=False, # Set to True if you want a logarithmic x-axis
    size_max=30, # Maximum size of the markers (reduced from 60)
    title='Food Cost vs. Taxes by State (Size: Childcare Cost, Color:
isMetro)',
    labels={
        'food_cost': 'Mean Food Cost',
        'taxes': 'Mean Taxes',
        'childcare_cost': 'Mean Childcare Cost',
        'predominant_isMetro': 'Predominantly Metropolitan'
    },

```

```
    template='plotly_white'  
)  
  
fig.show()
```

13. Gemini used for plot creation.

```
import plotly.express as px  
  
fig = px.scatter(  
    df_summary_metro,  
    x='housing_cost',  
    y='taxes',  
    size='transportation_cost',  
    color='state', # Color based on the state  
    hover_name='state',  
    log_x=False,  
    size_max=20, # Reduced bubble size for better differentiation  
    title='Housing Cost vs. Taxes by State (Size: Transportation Cost,  
Color: State)',  
    labels={  
        'housing_cost': 'Mean Housing Cost',  
        'taxes': 'Mean Taxes',  
        'transportation_cost': 'Mean Transportation Cost'  
    },  
    template='plotly_white'  
)  
  
fig.show()
```

14. Gemini used for plot creation.

```
import plotly.express as px  
  
fig = px.scatter_3d(  
    df_summary_metro,  
    x='transportation_cost',  
    y='healthcare_cost',  
    z='childcare_cost',  
    color='state', # Color by state  
    hover_name='state',
```

```

    title='3D Plot of Transportation, Healthcare, and Childcare Costs by
State',
    labels={
        'transportation_cost': 'Mean Transportation Cost',
        'healthcare_cost': 'Mean Healthcare Cost',
        'childcare_cost': 'Mean Childcare Cost'
    },
    template='plotly_white'
)

fig.show()

```

15. Gemini used for plot creation.

```

import plotly.express as px

fig = px.scatter_3d(
    df_summary_metro,
    x='childcare_cost',
    y='taxes',
    z='median_family_income',
    color='state', # Color by state
    hover_name='state',
    title='3D Plot of Childcare Cost, Taxes, and Median Family Income by
State',
    labels={
        'childcare_cost': 'Mean Childcare Cost',
        'taxes': 'Mean Taxes',
        'median_family_income': 'Mean Median Family Income'
    },
    template='plotly_white'
)

fig.show()

```

16. Gemini used for the PCA formula.

```

from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
import pandas as pd

# Select the relevant numerical columns for PCA from df_summary_metro

```

```

# The 'cost_columns' list already contains the desired columns
pca_data = df_summary_metro[cost_columns]

# Standardize the data before applying PCA
scaler = StandardScaler()
scaled_data = scaler.fit_transform(pca_data)

# Apply PCA with 3 components
pca = PCA(n_components=3)
pca_components = pca.fit_transform(scaled_data)

# Create a DataFrame for the PCA components
pca_df = pd.DataFrame(data=pca_components, columns=['PC1', 'PC2', 'PC3'])

# Add the 'state' column back to the PCA DataFrame for context
pca_df['state'] = df_summary_metro['state']

print("Explained variance ratio by each principal component:")
print(pca.explained_variance_ratio_)
print(f"Total explained variance: {pca.explained_variance_ratio_.sum():.2f}")

print("\nFirst 5 rows of the PCA components DataFrame:")
display(pca_df.head())

```

17. Gemini used for the NMF formula.

```

from sklearn.decomposition import NMF
from sklearn.preprocessing import StandardScaler
import pandas as pd

# Select the relevant numerical columns for NMF
nmf_data = df_summary_metro[cost_columns]

# Standardize the data before applying NMF
# NMF works best with non-negative data. If data contains negatives,
# consider scaling to a [0,1] range or using a different method.
# For cost data, values are inherently non-negative, but scaling can help.
scaler = StandardScaler()
scaled_data_nmf = scaler.fit_transform(nmf_data)

```



```

# Ensure data is non-negative for NMF. If StandardScaler produces negative
values,
# a common practice is to shift them to be non-negative, or use a min-max
scaler.
# Given our data are costs, they are already non-negative. StandardScaler
centers them,
# so we might end up with negative values. For a proper NMF application on
cost data,
# usually, you'd apply NMF directly to the non-negative raw data or
scaled-to-positive data.
# For demonstration, we will proceed with the StandardScaler output, which
may require
# careful interpretation or a shift if NMF algorithm has issues with
negatives.
# A common workaround if scaled_data_nmf has negatives is: scaled_data_nmf
= scaled_data_nmf - scaled_data_nmf.min()

# Apply NMF with 3 components
nmf = NMF(n_components=3, init='random', random_state=42, max_iter=1000)
nmf_components = nmf.fit_transform(nmf_data) # Apply NMF directly to
non-negative original data for better interpretation of components

# Create a DataFrame for the NMF components
nmf_df = pd.DataFrame(data=nmf_components, columns=['NMF1', 'NMF2',
'NMF3'])

# Add the 'state' column back to the NMF DataFrame for context
nmf_df['state'] = df_summary_metro['state']

print("First 5 rows of the NMF components DataFrame:")
display(nmf_df.head())

print("\nOriginal features' weights in NMF components (components_
attribute of NMF):")
display(pd.DataFrame(nmf.components_, columns=cost_columns, index=['NMF1',
'NMF2', 'NMF3']))

```